higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

WELDING
NQF LEVEL 4

(6031034)

22 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Use BLUE or BLACK ink only.
6. Write neatly and legibly.

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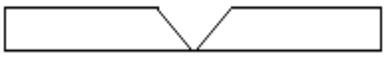
QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 The most common joint found in pipes is… joints.

A butt
B fillet
C edge
D lap

1.1.2 Name the type of groove shown below that is commonly used to prepare the edges for butt welding.



A Bevel-groove
B V-groove
C J-groove
D U-groove

1.1.3 Name the THREE phases of gas backing.

A Primary, tertiary and welding
B Primary, secondary and welding
C Secondary, final and primary
D Secondary, tertiary and primary

1.1.4 Welding leads are insulated … conductors.

A copper
B lead
C brass
D iron

1.1.5 Stainless steel contains … which makes it corrosion-resistant.

A cast iron
B nickel
C aluminium
D chromium

(5 × 1) (5)

1.2 List FIVE properties of copper alloys. (5)

1.3 Explain the set-up of a Shielded Metal-arc Welding (SMAW) process equipment. (5)

1.4 List FIVE factors that influence the size of electrodes used for Shielded Metal-arc Welding (SMAW). (5)

[20]

QUESTION 2

2.1 Direct-current Earth Negative (DCEN) can be used for Shielded Metal-arc Welding (SMAW) because more heat is concentrated in the electrode.

Draw a neat label sketch indicating the Direct Current Earth Negative (DCEN) set-up. (5)

2.2 Explain the chemical process that occurs during Shielded Metal-arc Welding (SMAW). (5)

2.3 List FIVE pre-checks that should be carried out on a pipe before it can be welded using Shielded Metal-arc Welding (SMAW). (5)

2.4 Give FIVE reasons why it is important to clean a pipe after welding it with Shielded Metal-arc Welding (SMAW). (5)

[20]

QUESTION 3

3.1 Explain the Gas Tungsten-arc Welding (GTAW) process. (4)

3.2 FIGURE 1 below indicates the equipment that constitute the Gas Tungsten-arc Welding (GTAW) set-up. Label items (A–F) in the ANSWER BOOK.

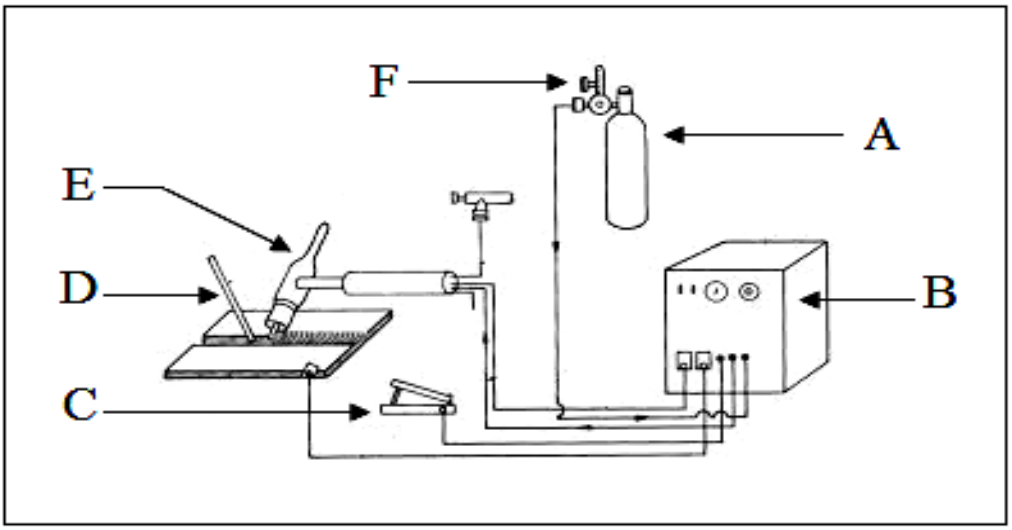


FIGURE 1 (6)

3.3 Choose a description from COLUMN B that matches an item in COLUMN A. Write only the letter (A–K) next to the question number (3.3.1–3.3.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.3.1	A collet	A	directs inert shielding gas over the welded area
3.3.2	The gas nozzle	B	noble gas
3.3.3	Travel angle of a Gas Tungsten-arc Welding torch	C	used to measure the strength of electric current
3.3.4	Tungsten	D	secures the electrode to the torch
3.3.5	A weld pool	E	a strong metal even at high temperatures
3.3.6	Slag	F	prevents the weld area and arc from reacting with the atmosphere
3.3.7	An inert gas is also known as	G	18° to 22° C
3.3.8	Amperage	H	flow of electrons from the negative to the positive pole
3.3.9	Polarity	I	waste matter that comes off metal during melting
3.3.10	Shielding gas	J	15° to 20°
		K	the molten metal created during welding

(10 × 1) (10)
[20]

QUESTION 4

4.1 Name TWO differences between a Gas Tungsten-arc Welding (GTAW) filler rod and a Shielded Metal-Arc Welding (SMAW) electrode. (2 × 2) (4)

4.2 List FIVE causes of 'Incomplete Fusion' during the Gas Tungsten-Arc Welding (GTAW) process. (5)

4.3 Explain how the 'Overhead Weld Position' is carried out during Gas Tungsten-Arc Welding (GTAW). (5)

4.4 Draw a Tungsten Inert Gas (TIG) welding gun that is used in the Gas Tungsten-Arc Welding process.

Indicate the nozzle, collet body, tungsten electrode, collet, cap and torch body. (6)

(4 × 5) **[20]**

QUESTION 5

5.1 Before MIG welding can begin, all tools and equipment must be prepared.

Describe how you would select or prepare the equipment for Metal Inert-gas (MIG) welding. (5)

5.2 Explain why the aluminium oxide coating on aluminium needs to be removed before MIG welding can take place. (5)

5.3 Name TWO methods that can be used to clean a gas nozzle on the MIG gun. (2)

5.4 State THREE functions of the Metal Inert Gas (MIG) wire feeder gun (3)

5.5 FIGURE 2 indicates the different types of welding symbols.

List FIVE types of information that can be identified from the welding symbol below.

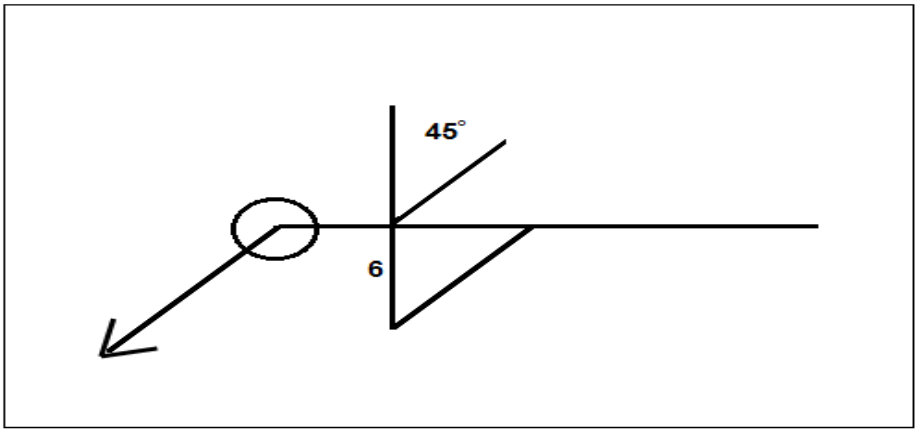


FIGURE 2

(5)
[20]

TOTAL 100

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